DropBox :

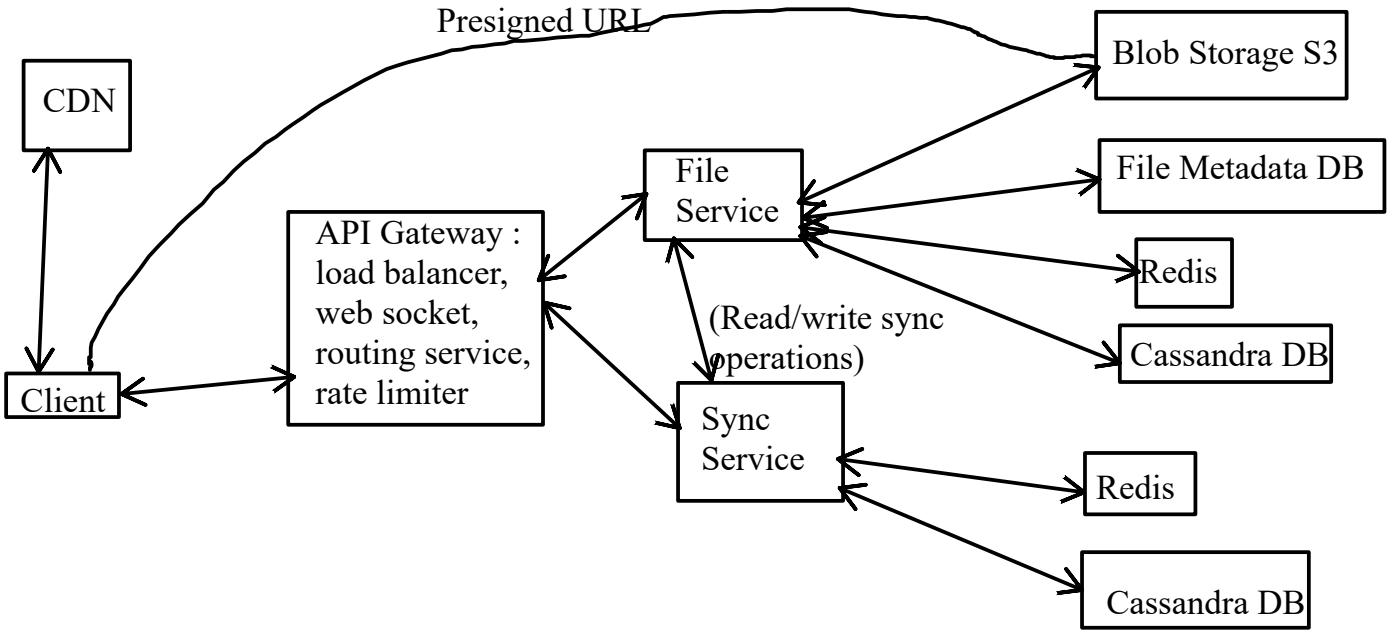
Functional requirements:
user should be able to upload/download any file based on his preference,
user should be able to search for any file based on his preference,
user should be able to upload any file>50GB based on his preference,

Non-Functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
file,
filemetadata,
history

API :
POST:/upload/post
{
userID,
{filemetadata}
}
GET:/search/{fileID}

HLD :



1) we need to chunk 50GB file into multiple chunks
2) we need to fingerprint or hash each piece of these chunks separately
3) we need to upload each piece of these chunks separately
4) update completed status for each piece of these chunks separately